Auditing a Streaming Subscription Churn Model

RESPONSIBLE DATA SCIENCE
FINAL PROJECT

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Group 87

Background

Overview

Welcome to the DSC 783 Final Project.

For this project, you are a Business Analyst working for a Music Streaming Service called PlaylistPro.

At PlaylistPro, one of company's top priorities is to **reduce customer churn (stop customers from cancelling their subscriptions).**

To reduce churn, PlaylistPro plans to reduce the cost of the music streaming service for customers that are likely to cancel their subscription. But, the company currently does not know which customers are likely or unlikely to cancel their subscription.

Thus, PlaylistPro would like you to build a Supervised Classification Model to predict if a customer will churn based on their subscription information and listening habits. To help you build a model, the company's Operations team created a dataset of 125,000 customers to train your model on. More details are below:

CLICK HERE TO VIEW THE KAGGLE COMPETITION

PURPOSE

- **The System**: PyTorch MLP Neural Network.
- **The Task**: Predict subscriber churn for PlaylistPro based on demographics and listening habits.
- **The Intervention**: Offer financial retention discounts to "at-risk" users.

CONCERNS

- **High Precision** (Too conservative) → Miss churners, unfarily exclude them from discounts
- **High Recall** (Too liberal) → Flag everyone; wastes company's marketing budget

Input & Output

INPUT

- SYNTHETIC DATASET SIMULATING STREAMING PLATFORM USER DATA, 1 ROW = 1 USER

- NOTABLE INPUTS FROM KAGGLE COMPETITION "STREAMING SUBSCRIPTION CHURN MODEL."
  - CUSTOMER_ID (INT)
  - AGE (INT)
  - SONG_SKIP_RATE (FLOAT)
  - AVERAGE_SESSION_LENGTH (FLOAT)
  - SUBSCRIPTION_TYPE (STR)
  - PAYMENT_PLAN (STR)
  - WEEKLY_HOURS (INT)
  - CUSTOMER_SERVICE_INQUIRIES (STR)

- *CUSTOMER_SERVICE_INQUIRIES* TENDS TO CORRELATE POSITIVELY WITH CHURN WHILE *WEEKLY_HOURS* CORRELATES NEGATIVELY

OUTPUT

- BINARY CLASS LABEL:

LABELLED CHURNED (1) → DISCOUNTS
LABELLED NOT CHURNED (0) → NO DISCOUNTS

- UNDER THE HOOD, THE PYTORCH MLP OUTPUTS A SIGMOID PROBABILITY BETWEEN 0 AND 1, THRESHOLDED AT 0.5.

Detailed Analysis Plan

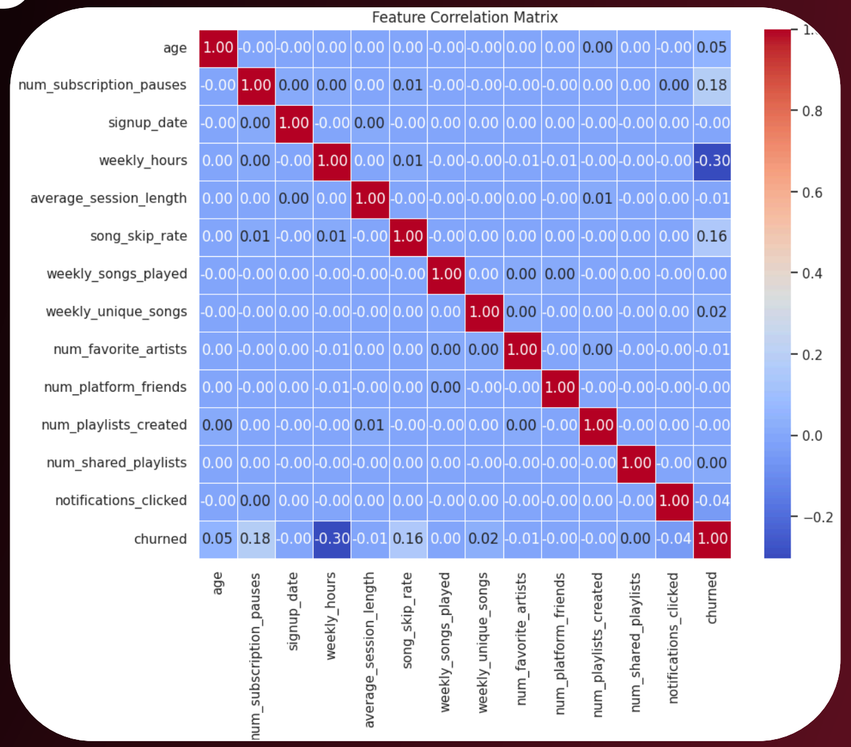
1. DEFINE AGE SUBGROUPS
2. COMPUTE PRECSION, RECALL, FPR, FNR*
3. EVALUATE DEMOGRAPHIC PARITY, EQUALIZED ODDS, PREDICTIVE PARITY
4. SHAP FOR FEATURE CONTRIBUTION
5. COMPUTE DOWNSTREAM DOLLAR IMPACT

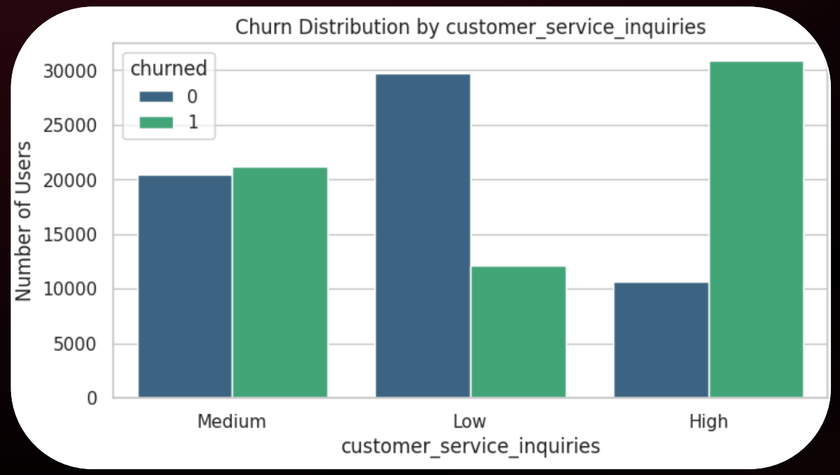
Based on findings we'll recommend threshold adjustments, feature removal, or calibration strategies to improve fairness without unacceptably hurting accuracy.

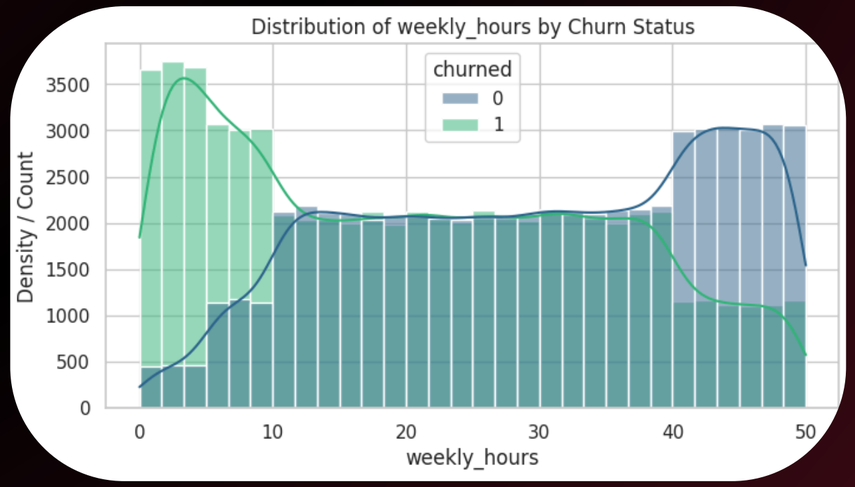
* special attention to FNR since those represent churners who are denied retention discounts

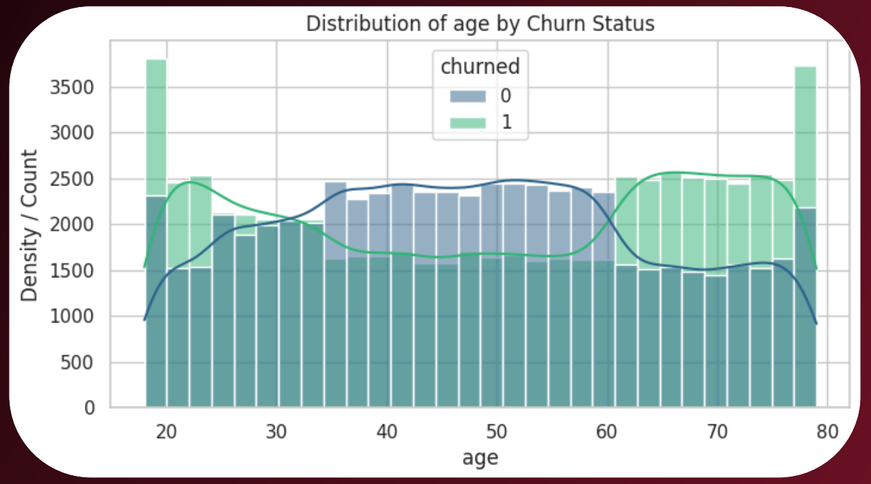
Early Takeaways

- MODEL'S STRONG AGGREGATE ACCURACY (~84–87%) LIKELY MASKS UNEVEN ERROR RATES ACROSS DEMOGRAPHIC GROUPS
- BEHAVIOR DRIVES THE MODEL
  - BEHAVIOR AS A PROXY TO DISCRIMINATE BASED ON AGE
- IF THE MODEL HAS A HIGHER FALSE NEGATIVE RATE FOR SPECIFIC AGE GROUPS, THOSE USERS ARE BEING SYSTEMATICALLY EXCLUDED FROM RECEIVING FINANCIAL RETENTION DISCOUNTS









Thank you

Questions?